Al- Balqa' Applied University
Faculty of Engineering Technology
Basic Science Department
Department of Physics and Basic Sciences

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم الفيزياء والعلوم الأساسية

24

Midterm Exam Phy-111C

Time of exam: 60 minutes

[redacted] موعد المختبر : [redacted] مدرس المادة : [redacted] الاسم بالعربية [redacted]

Note: When you need consider the acceleration of gravity (g) =10 m/s²

Q (1)	1	2	3	4	5	6	7	8	9	10
	B	B	D	C	D	C	A	D	B	B

Q(2)	1		2		3		4		5	
	T	F	T	F	T	F	T	F	T	F

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Q.1: Circle the correct answers, and then move these answers to the table :(20 marks)

1) The instrument which used to take this reading (2.265cm) is:

A) ruler	B) micrometer	C) Vernier caliper	D) none of all

2) For an (3) forces equilibrium system as shown in the table,

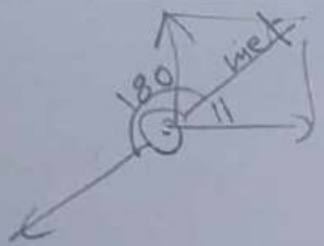
The magnitude of (F_3)

A) 0.99 N	B) 1.06 N	C) 0.92 N	D) 9. N

mass	force	direction	x-com	y-com
80 g	F1	?	0.8	0
70 g	F2	?	0	0.7
M3	F3	?	?	?

3) The direction of (F_3):

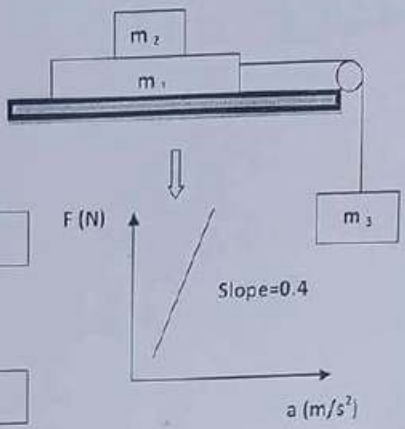
A) 229.4°	B) 49.4°	C) 221.2°	D) 41.2°



4) Using significant figures rules (8.58X3.6) =

A) 30.9	B) 30.6	C) 31.	D) 30.60

In Newton's second law experiment, If m_1=0.35 and m_2=10g. (kg)



Answer question (5) and (6)

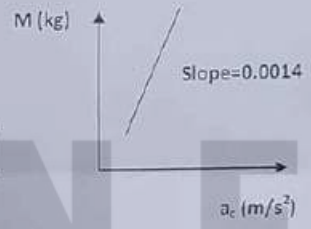
5) The total force which moves these masses is:

A) 0.1 N	B) 0.35 N	C) 0.3 N	D) 0.4 N

6) The magnitude of (m_3) in kg, is:

A) 0.04	B) 0.03	C) 0.035	D) 0.01

In uniform circular motion experiment, if you got the graph shown,



Answer question (7) and (8)

7) The mass which is rotate in uniform circular motion (m) in (kg) is:

A) 0.14	B) 0.014	C) 0.1	D) 0.2

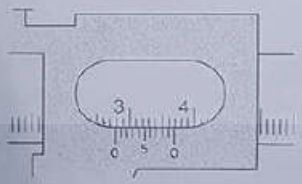
8) When ac= 295.78 m/s², and (r=30cm).the time of (20) revolutions is:

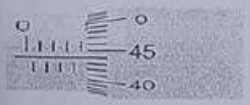
A) 6.67 s	B) 8.67 s	C) 9.2 s	D) 4 s

9) Using significant figures rules, (74025000 ÷ 987) is:

A) 0.8X10⁵	B) 0.750X10⁵	C) 0.75X10⁵	D) 75000

10) The sum of micrometer and caliper reading in (mm) is:
(using (s.f) rules)





A) 32.24	B) 32.2	C) 33.	D) 33.1

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Q (2): Put (T) in front of the correct phrase, and (F) In front of the wrong phrase. Then move these answers to the table. (10marks)

1) The relation between (a) and (1/m), when the force is constant.is linear directly proportional. ( ). T

2) The benefit of crocodile clip in uniform circular motion is to keep the radius constant when we rotate the small mass (m). (T).

3) In Newton's second law experiment, you can calculate always (V_f), as $V_f=1/t^2$.

(F).

4) In Newton's second law experiment, the decline of air track acts on results. (T).

5) The uncertainty for the area of circle, is $\Delta A=(2\Delta r/r)A$. (T).

πr^2 $\Delta A \frac{2\Delta}{r}$

Good luck for all

$Mg = \frac{v^2}{r}$ $4\pi r$

$Mg = 4\pi^2 r m f^2$

$\frac{Mg}{10} = \frac{14}{10}$

$\frac{18}{100}$ 1,4

4 x

$mac = \frac{1.6}{t^2}$

295.78

27,7

4,44 mm

0,4 = ,36 + m_3

↓ (a) $\alpha \frac{1}{m}$ ↑

$m_3 g =$

.936

.,01

$m_2 g - T = 3.6$

$ma_c = 4\pi^2 r m f^2$

$295.78 = \frac{11.8}{t^2}$

.104

102

T -

5.6

$\Sigma F_y = T$

$\Sigma F_y = 3.6$

$T - m_3 g = 3.6$

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12

Midterm Exam Phy-111(A)

[redacted] | [redacted] | [redacted]

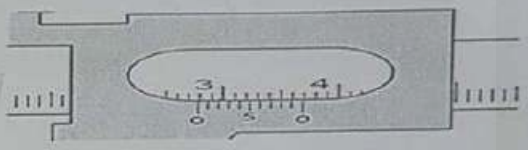
الاسم بالعربية : [redacted] | مدرس المادة [redacted] | عد الطلاب : [redacted]

1+7

• *Q 1: Fill the blank with suitable answer :* *(13.5 marks)*

1) If we turned a closed micrometer (5) turns and (1/5) of a turn, then the micrometer reading is ([redacted] mm).

5.20 mm
= 2.6 mm



2) L= ([redacted] mm) ± ΔL= (0.005 mm) 0.05 mm

2.88 27.8mm

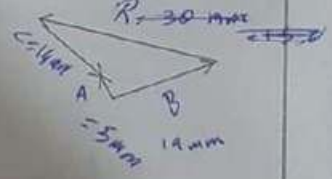


3) If the diameter for a ball was measured using a micrometer as shown then: ΔA= (± 0.16 mm^2). 0.133 mm^2

12.1

4) If the scale used to draw this vector diagram for three forces was (2mm = 1 N),then the magnitude of the net force is(12 N)

R = 4mm → 2 N



5) The instrument used to take this reading (13.3 cm) is (Vernier caliper, Micrometer, Ruler).

(Ruler).

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6) Using significant rules to find 306000/45.0 = ($680 * 10^1$).

7) In gravitational potential energy experiment, before start move toward down from (h_i=100cm) the mass was (m=20g) and (M=200g) ,then the kinetic energy when (m) reach (h=40cm) is: (joule).

8) For the above question the final velocity (V_f) when m reach ground is: (m/s).

9) Using significant rules to find , 4.207÷3.78= (7.99

- *Q2 : Circle the correct answer:(10.5 marks)*

1) If a student used a ruler with a maximum range (15 cm) to measure the length of a table which has length (1.8 m), then the uncertainty in length (ΔL) is:

a)3mm b)4.5mm c)8.5mm d) 6mm

2) If the relation between A&B given as (C =A-B), if A = 5.28 and B=1.8, then (C±ΔC) is:

(Using significant rules)

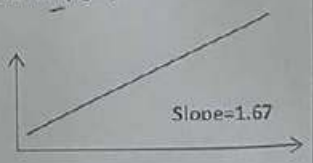
a) 3.48±0.05 b) 3.5±0.1 c) 3.5±0.06 d) 3.48±0.055

3) F1,F2,F3 are three forces in equilibrium ,if you know that (F_{3x} =-1.32 N , F_{3y} =-7.5 N),the direction of resultant between (F1,& F2) according to x-positive axis is :

a) 100 b) 80 c) 280 d) 260

4) In Newton's second law experiment, if we got this graph,

then the constant hanged mass in the experiment is:



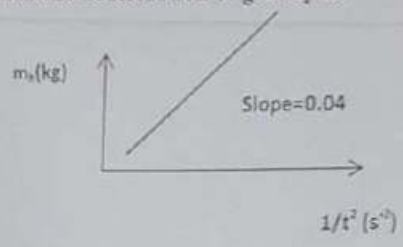
a) 15g b) 25g

c) 60g d) 80g

5) The graph shown below is from Newton's second law experiment, If the distance that (M_w) moved in all trials was (60 cm) and total mass (300g) , then the acceleration of gravity is:

a) 8 m/s^2 b) 7 m/s^2

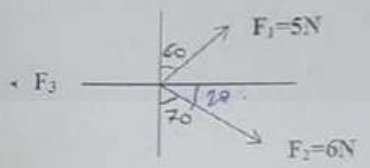
c) 9 m/s^2 d) 12 m/s^2



6) For equilibrium state, the magnitude of(F3)is:

a) 9.5N b) 9.98 N

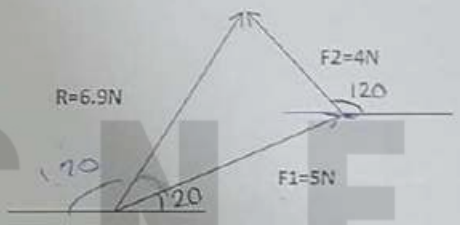
c) 8.45N d)10.8N



7) The direction of (R), according to x-positive axis'sis:

a) 82.8° b) 96.3°

c) 61.5° d) 54.8°



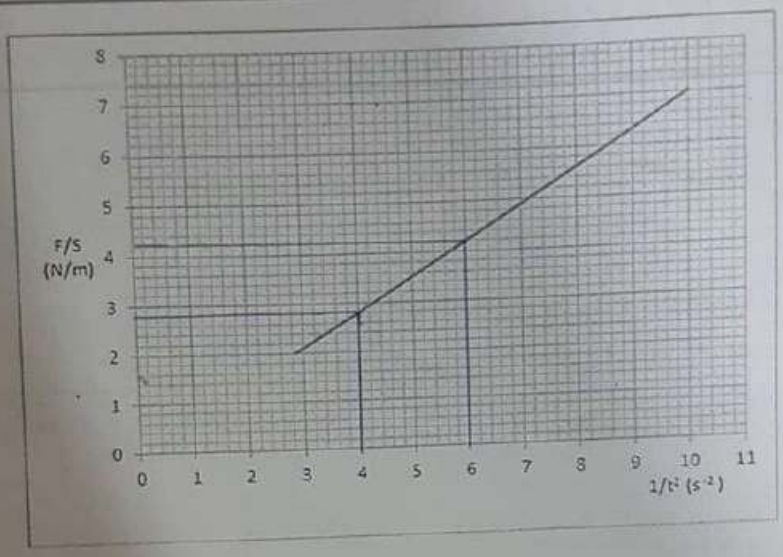
- *Q3 :*In Newton's second law experiment, the distance that (M_w) was moved in all trials considered to be (80cm). then use the graph to answer the following :

(2 marks)a) The physical meaning of the slope is (total mass (m))

(2 marks) b) The total mass is (0.2 kg)

(2 marks) c) If (t=0.4s) the hanged mass (m_h) will be (4 kg)

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Good luck

AL-BALQA' APPLIED UNIVERSITY
Faculty of Engineering Technology
Physics Department

Fall 2016
Date: April 7, 2016

Midterm
Physics 111

Name: [redacted]
Consider g = 10 m/s² %
Instructor: [redacted]

1) Circle the correct the best answer for the following questions.

Q.1 A particle of mass 5 Kg is just starting to move when a force of 30 N is applied horizontally. What is the coefficient of static friction μ_s?
a) 0.3 b) 1.5 c) 0.7 d) 0.5 e) 0.6

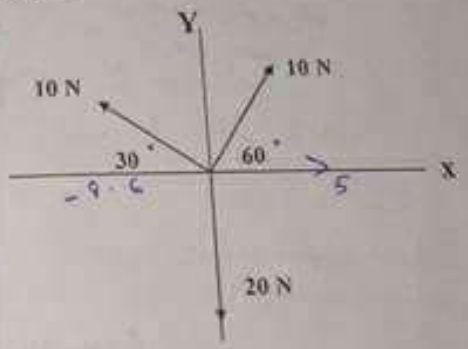
Q2 A student wanted to calculate the density (ρ) of Mass (M). If $M \pm \Delta M = (10 \pm 0.01)$ gram and its volume $V \pm \Delta V = (5 \pm 0.1)\ cm^3$, the absolute error of the density (Δρ) in g/cm^3 is:
a) 0.08 b) 0.04 c) 0.06 d) 0.02 e) none

Q.3 A Vernier caliper was used to measure the length of a cube. What is the error in length (ΔL) ?
a) 0.50 mm b) 0.05 mm c) 0.5 mm d) 0.005mm e) 0.0005 mm

Q.4 Which of the following measurements represents the reading of the micrometer?
a) 9.25 mm b) 8.0 cm c) 1.50 cm d) 2 mm e) 10.1 mm

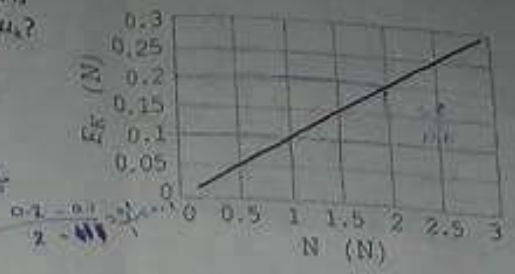
Q.5 What is the value of 0.484 × 0.080 to the appropriate number of significant figures?
a) 0.04 b) 0.03 c) 0.039 d) 0.0387 e) 0.038

Q.6 In the Figure shown.
What is the magnitude of the resultant force (in N) along x axis?
a) 13.66 b) 5.00 c) 8.66 d) 3.66 e) 0



Q.7 From the graph shown. What is the coefficient of kinetic friction μ_k?

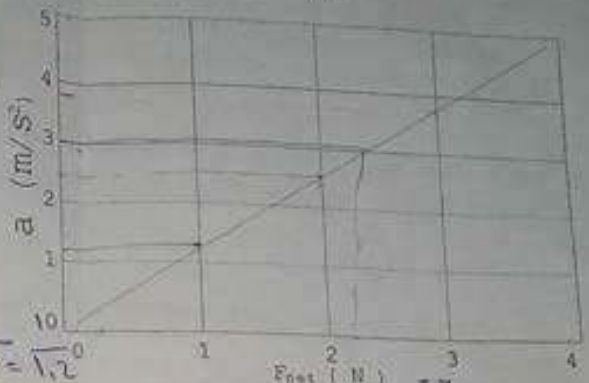
(a) 0.1 b) 0.3 c) 0.4 d) 0.5
e) 0.7



2

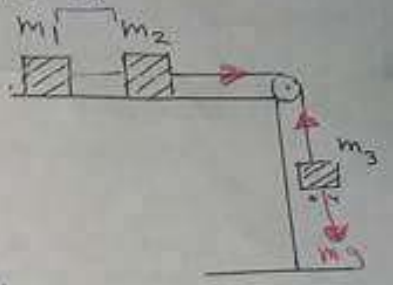
In the Newton's second law experiment, we add a new mass m_3, connected to m_2 as shown in the figure. (Show your calculations and write the final answer in the box).

a) What is the sum of the three masses.



Σm = Slope $= \frac{\Delta a}{\Delta a} = \frac{2-1}{2.6-1.4} = 1.2$

0.8 Kg



b) Find F_{net} when $m_3 = 0.4$ Kg

$m_1 + m_2 = m_3$

$m_1 + m_2 = 0.4$

$F = mg = 0.4 \times 9.8 = 3.92$ N

3.92 N

c) What is the acceleration of the system when $m_3 = 0.3$ Kg?

$F = ma$

$\frac{3.92}{0.3} = \frac{0.3a}{0.3}$ $\quad a = 13.06$

13.0 m/s²

$a = 3.75$ m/s²

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Physics 111
Midterm Exam (a)
Time: 50 minutes

[redacted] موعد المختبر :

[redacted] مدرس المختبر

[redacted] الاسم بالعربية

Circle the correct answer

1. The number of significant figures in the measurement (L = 75.00 m) is:

a) 1 b) 2 c) 3 d) 4

2. The addition of the measurements (6.201 cm, 7.4 cm, 0.68 cm, and 12.0 cm) equals

a) 26.281 cm b) 26.3 cm c) 26.28 cm d) 26.4 cm

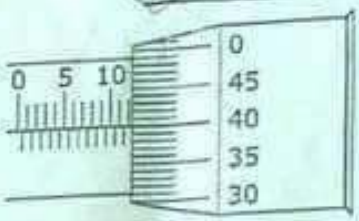
3. The multiplication of the measurements (3.2145 km × 4.23 km) equals

a) 13.6 km^2 b) 13.597335 km^2 c) 13.59 km^2 d) 13.60 km^2

4. Which of the following measurements represent the reading of the micrometer?

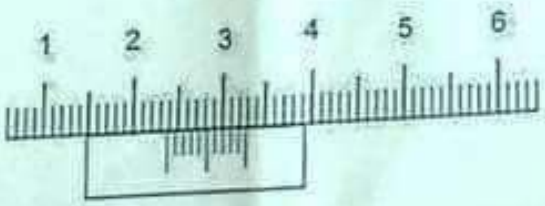
a) 10.3 mm b) 8.0 cm c) 1.50 cm d) 2.32 mm

5. The reading of the micrometer is:



a) 12.90 mm b) 10.240 mm
c) 12.40 mm d) 10.60 mm

6. The reading of the Vernier caliper is:

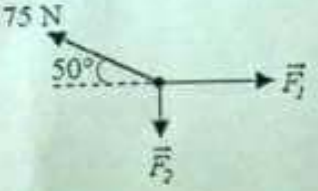


a) 2.30 cm b) 2.38 cm
c) 1.48 cm d) 2.48 cm

7. A cylinder has a radius of $R \pm \Delta R = (2.00 \pm 0.01)$ cm and a length of $L \pm \Delta L = (10.0 \pm 0.1)$ cm. The error in the volume ΔV is:

a) 0.02 cm^3 b) 0.015 cm^3 c) 2.51 cm^3 d) 1.9 cm^3

8. A stationary object has three forces acting on it, as shown in the diagram, What are the magnitude of $\vec{F}_1$?



a) 57.5 N b) 48.2 N c) 15 N d) -20 N

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الاسم ..... مدرس المادة: ..... موعد المختبر: 2 - 5 الإثنين		

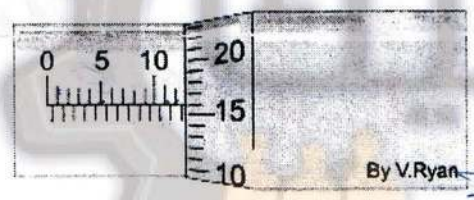
Q1: Determine the number of significant figures in the following measured values:

1- 2.3 cm ..... 2 numbers .....
2- 3.589 s ..... 4 numbers .....
3- 4.56×10^{-6} kg ..... 3 numbers .....
4- 1.0003 J ..... 5 numbers .....

Q2: Carry out the following arithmetic operations :

1- The sum of of the measured values 756, 37.2, 0.83 and 2.5 = ... 797 .....
2- The product 0.0032× 356.3 = ... 1.1 .....
3- The division $\frac{1.05}{5.346}$ = ... 0.196 .....
4- The subtraction 0.082 - 1.02 = ... -0.94 .....

Q3- Read the scale of the micrometer, what is the error of the reading



125.16 ..... 12.66 mm .....

Q4: Vector Experiment

If you have three forces, one of A(m_1=0.200 Kg)pointing at 30^0, the second B(m_2=0.200 Kg)pointing at 120^0, and the third C as shown in the force table. If the ring remains centered around the pin without being moved in any direction, complete the following Table

Vector	Mass, kg	Magnitude in (N)	Direction	x-component
A	0.200	2	30^0	$2\times\cos 30 = 2\times 0.86 = 2$
B	0.200	2	120^0	1 in negative direction of (x)
C	0.300	3	72^0	1

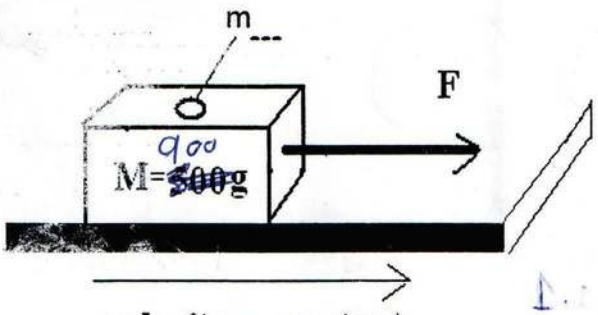
$\vec{A} = 1.72\,i + 1\,j$

$\vec{B} = -i + 1.72\,j$

$i + 3j$

1.0×0.200

Q5- Friction Experiment: If the applied force F moves the block M and the mass m with constant velocity, find the kinetic coefficient of friction in each case and its average value.



μ_k	m(kg)	F(N)
0.61	0.100	6.00
0.63	0.200	6.80
0.62	0.300	7.30

μ_k average= 0.62

Q6- Measurements

Ahmad measured the radius and the mass of a sphere using the micrometer and the balance, R= 0.86 mm and m= 4.3 g,

Calculate the volume(V±ΔV) and the density ((ρ±Δρ)) and the error in each of them.

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Department of Applied Science
B

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15/20

Physics 111
Midterm Exam
Time:50 minutes

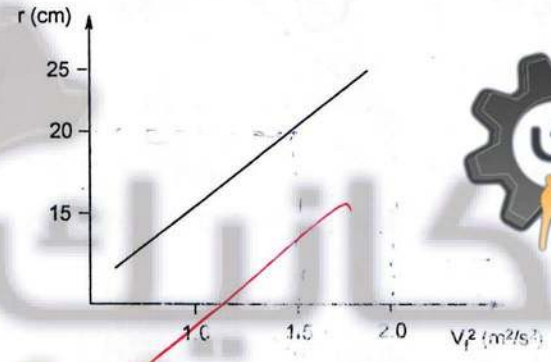
الاسم بالعربية:	الشعبة: الاثنين	المدرس: حلوه طه

Circular motion

1) A man is moved from rest around circular path with radius 60 m and he made 10 turns in 10 minute. His final velocity in **(m/s)** is

a) 7.00 b) 6.28 c) 4.19 d) 3.50

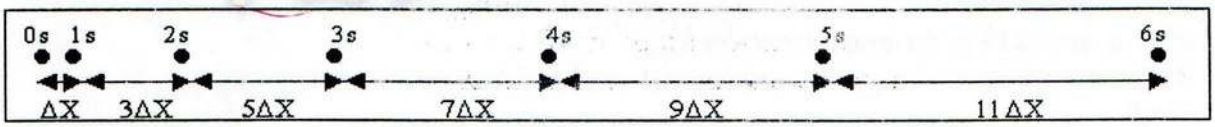
2) From the graph, the experimental centripetal acceleration is

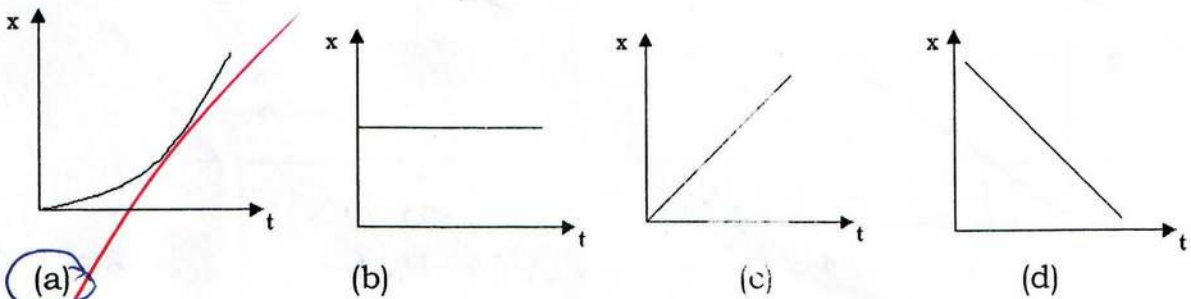


a) 0.1 m/s² b) 10 m/s² c) 15 m/s² d) 0.15 m/s²

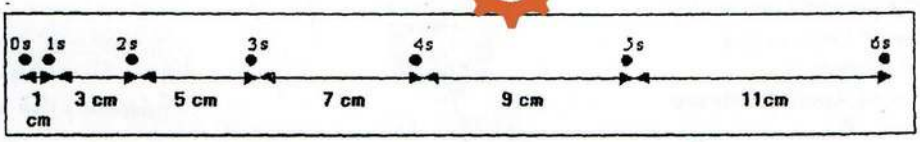
Kinematics of rectilinear motion

3) For the ticker tape shown below, the correct graph representing position versus time is.





(a) (b) (c) (d)

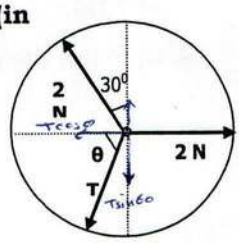


4) For the ticker tap shown above, the average velocity (in **cm/s**) for the period **t = 1s** to **t = 4s** is:

(a) 3 (b) 2.5 (c) 5 (d) 3.5

Vector: Force table

5) For the experimental setup shown. If the body located, at the center, and its at equilibrium. The values of **T (in N)**, and **θ (in degree)** are respectively:

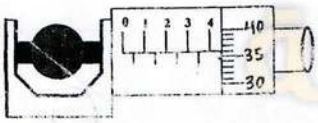


a) 2.64, 71.9 b) 3.36, 68.9 c) 1.98, 59.83

d) 3.36, 79.1 e) 2.64, 79.1

Measurments and uncertainities

6) A micrometer is used to measure the diameter of a sphere as shown in the figure.



The volume of the sphere in (**mm³**) is:

a) 37.7 b) 347

c) 43.4 d) 302

7) A Vernier caliper was used to measure the length of a cube, the error in length ΔL is

a) 1mm b) 0.5mm c) 0.05mm d) 0.1mm

,,,, Experimental setup shown in Fig.1, when the masses M_a+M_h were kept constant and equal to **210 g**, the graph between $M_h g$ the hanging mass times gravity acceleration, and **a,** is the acceleration is shown in Fig.2.

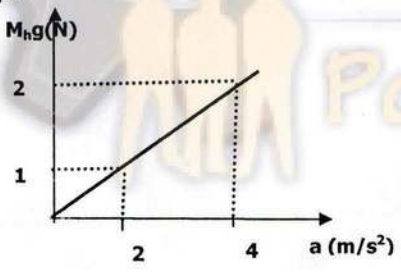


Fig.2

(Note: g= 9.8 m/s²)

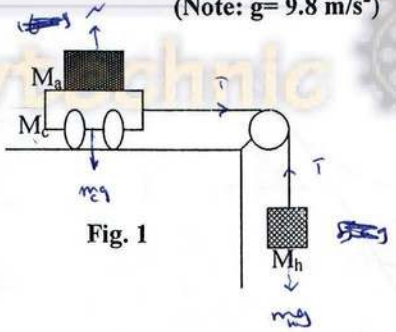


Fig. 1

Q8) The mass of the cart M_c (in gm)is : _____

Q9) If the added mass is M_a=120 g,the magnitude of the **acceleration** for this system (**in m/s²**) is:

$F = ma$

$F = (M_c + M_a)\,a$

$m_h = \frac{f}{a} = \frac{2}{4} = 0.5\ g$

$2 = (M_c + 209.5)4$

$0.5 = M_c + 209.5$

$M_c = 209$

2

$F = ma$

$a = \frac{F}{m}$

$a = \frac{0.5}{418} = 1.1962 \times 10^{-3}\ m/s^2$

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جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم التطبيقية

Physics 111
Midterm Exam
Time: 50 minutes

18.5/20

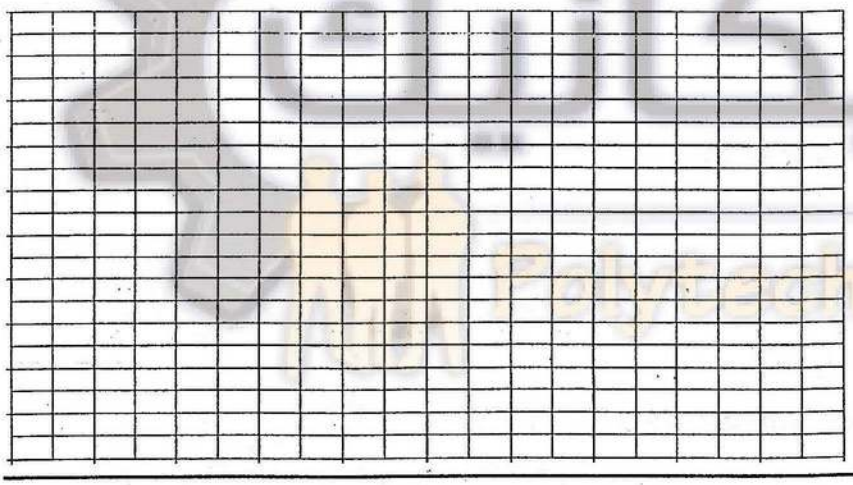
الاسم بالعربية: مدرس المختبر: الدكتور زياد عليمات موعد المختبر : الخميس (٢-٥)

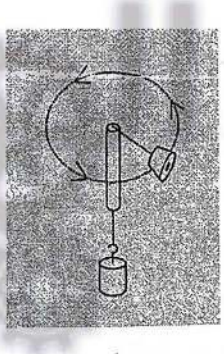
Q1: Uniform Circular Motion

a) In the circular motion experiment, a student studied the relation between the centripetal force and the square of the speed (v^2) for rotational body (m) by making the radius (r=0.45 m).

M (kg)	Period T (s)	F_c (N)	v(m/s)	v^2 $(m/s)^2$
0.02	1.15			
0.04	0.82			
0.06	0.67			

1. **Complete the table?**
2. **Plot a suitable graph between F_c and v^2**
3. **By using the slope of the graph, find the mass (m) of the rotational body.**





Q.2) A ball of mass 0.5 kg is attached to the end of a string 1.5 m long and whirled in a horizontal plane at constant speed. If the cord can withstand a maximum tension of 50 N, what is the maximum speed of the ball before the string breaks?

38.7 m/s

Q.2: Friction Force:

Q.1) For the inclined surface, if μk=μs=0.3; if the mass slides on the surface with constant velocity. Find the angle of the inclined surface?

$M_s = \tan\theta$
$0.3 = \tan\theta$
$\theta = \tan^{-1} 0.3$
$\theta = 16.7^\circ$

Q.2) the force F=4N, is applied to move the mass M=500 gram, on friction surface with constant velocity, find the kinetic coefficient of friction between the mass and the surface? [The gravity acceleration = 9.81 m/s²]

$\mu k = \frac{F_k}{N}$ ... $F - F_k = m.a$ ... $F - F_k = 0$ ... $\sum F = F_k$ ... $4 = \mu_k (m) g$

$4 = \mu_k * 500 * 10^{-3} * 9.81$ → $\mu_k = 0.815$

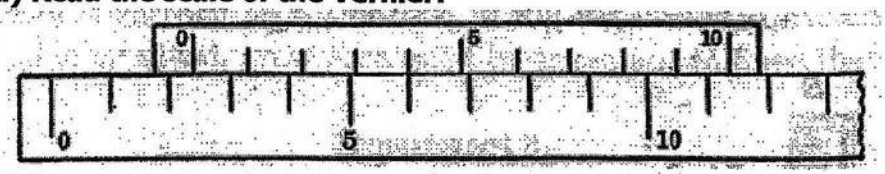
Q3: Vectors

Find the resultant force of the following two forces: F1 = 1.96 N (weight of 200 gram) at 60° from positive x axis. F2 = 2.94 N (weight of 300 gram) at 140° from the positive x axis.

The resultant force = 3.81 ...N, with direction = 109.5 ...degree

Q4: Measurements

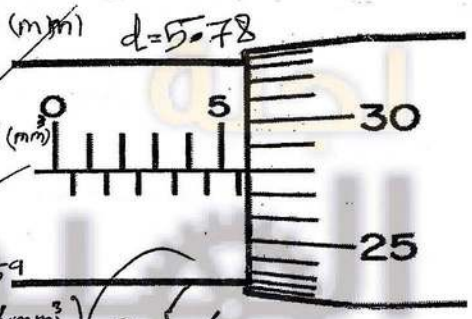
a) Read the scale of the Vernier:



2.4

b) In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was M ± ΔM = (2.1 ± 0.05) g. calculate:

a) The Radius of the sphere, R ± ΔR = 2.078 ± 0.005 (mm)



b) The volume of the sphere, V ± ΔV = 9.090 ± 0.0054 (mm³)

$\frac{4}{3}\pi r^3$ ... $3\frac{\Delta r}{r} = 3 * \frac{0.005}{2.78}$

c) The density of the sphere, ρ ± Δρ = 0.023 ± 0.00059 (g/mm³)

$\frac{M}{V}$ ... ΔM+ΔV

2.5/3

Q5: Significant Figures and Errors

a) Express the results of the following calculations to the appropriate number of significant figures?

1) 7.939 + 6.26 + 11.1 = 25.3

2) (27.2 x 15.63)/1.846 = 230.

b) Rounding the following numbers:

1) (0.07284) to 3 significant digits: 0.0728

2) (10.5660) to 4 significant digits: 10.57

3) (726.835) to 3 significant digits: 727

c) Determine the number of significant figures in the following numbers:

1) 005.00 x 10^{-4} cm ... 1

2.) 0.0350 gram ... 3

3.) 1.0003 Joul ... 5

4.) 50000 N ... 1

4/5

5/4

d) If X± ΔX = (4.00 ± 0.02) and $Z = \sqrt{X}$

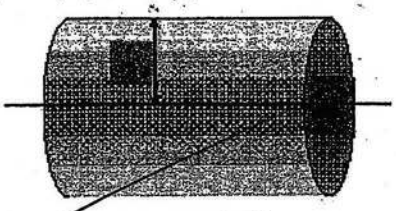
Calculate Z ± ΔZ = 2.00 ± 0.005

e) Experimental value of gravity is found to be 10.07 m/s². Accepted value of "g" is 9.81m/s². Calculate the percent error? .....

$0.027 * 100\%$ → 2.7%

2.65%

6. A long nonconducting cylinder of radius (R) has a positive charge of a uniform density (ρ) distributed throughout its volume. The magnitude of the electric field at distance r from the cylinder axis, where $r < R$ is given by:



a) $\frac{\rho r}{3\varepsilon_o}$ b) $\frac{\rho R^2}{2\varepsilon_o r}$ c) $\frac{\rho r^2}{2\varepsilon_o}$ d) $\frac{\rho r}{2\varepsilon_o}$ e) $\frac{\rho R^3}{2\varepsilon_o r^2}$

7. A uniform linear charge density of 4.0 nC/m is distributed along the entire x axis. Consider a spherical (radius = 5.0 cm) surface centered on the origin. The electric flux (in N m²/C through this surface is equal to:

a) 45 b) 68 c) 62 d) 79 e) 23

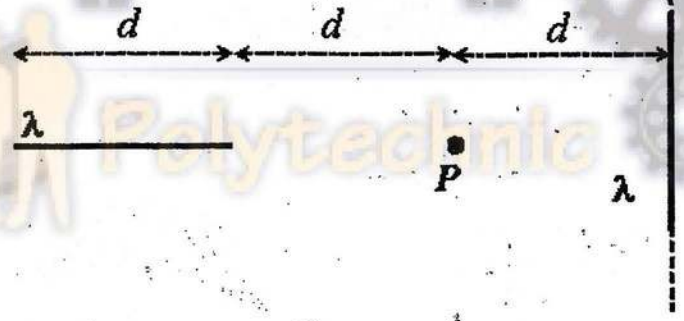
8. Positive charge Q is distributed uniformly throughout an insulating sphere of radius R, centered at the origin. A particle with positive charge Q is placed at x = 2R on the x axis. The magnitude of the electric field at x = R/2 on the x axis is:

a) $\frac{17Q}{72\pi\varepsilon_o R^2}$ b) $\frac{Q}{8\pi\varepsilon_o R^2}$ c) $\frac{Q}{72\pi\varepsilon_o R^2}$ d) $\frac{Q}{4\pi\varepsilon_o R^2}$

9. A charge of +80 μC is placed on the x-axis at x = 0. A second charge of -50 μC is placed on the x-axis at x = 50 cm. the magnitude of the electrostatic force on a third charge of 4 μC placed on the x-axis at x = 30 cm is equal to:

a) 45 N b) 77 N c) 25 N d) 13 N e) 39 N

10. In the figure shown two rods with uniform linear charge density (λ), the vertical rod is infinite and the horizontal rod is finite with a length (d), the magnitude of the electric field resulting from both rods at the point (P) is :



a) $3k\lambda/2d$ b) $2k\lambda/d$ c) $5k\lambda/2d$ d) $k\lambda/2d$

باقي اكل بالباقل

$\frac{2\lambda ke}{d} + ke\lambda\left(\frac{1}{2d}\right)$

$2 * \frac{2\lambda ke}{2d} + \frac{ke\lambda}{2d}$

$\frac{4\lambda ke + ke\lambda}{2d} = \boxed{\frac{5ke\lambda}{2d}}$

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جامعة البلقاء التطبيقية
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قسم العلوم التطبيقية

Physics 111
Midterm Exam
Time: 50 minutes (A)

الاسم بالعربية: مدرس المختبر: د. محمود ياسم موعد المختبر: ٢-٥ ع

Q1: Significant Figures and Errors

a) Express the results of the following calculations to the appropriate number of significant figures?

1) $0.424 \times 0.090 =$ 0.038

2) $\frac{1.36 \times 0.5}{2.6} =$ 0.3

4) $37.76 + 3.907 + 226.4 =$ 268.1

3) $3.76 + 14.83 + 2.1 =$ 20.7

4

b) **If** $X \pm \Delta X = (4.00 \pm 0.02)$ and $Z = \sqrt{X}$

Calculate $Z \pm \Delta Z =$ $2,00 \pm 0,005$

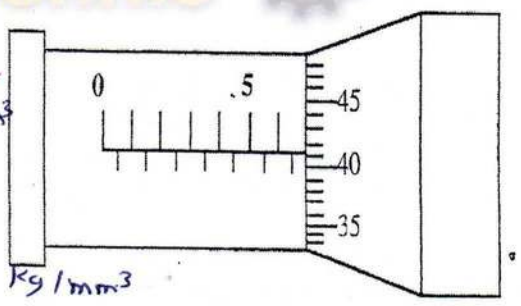
Q2: Measurements

In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was $M \pm \Delta M = (2.1 \pm 0.05)$ g. calculate:

a) The Radius of the sphere, $R \pm \Delta R =$ $\left(\frac{6,91 \pm 0,005}{2}\right)$ mm

3

b) The volume of the sphere, $V \pm \Delta V =$ 1382.04 ± 0.6 mm³



c) The density of the sphere, $\rho \pm \Delta\rho =$ 1.519 ± 0.636 kg/mm³

Q3: Vectors

Given three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$, if $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$, complete the following table:

6

g = 10

Force	Mass (kg)	Force (N)	Direction	x-component	y-component
$\vec{F}_1$	0,15 Kg	1,5 N	0°	$F_1 \cos 0 = 1,5$	$F_1 \sin 0 = 0$
$\vec{F}_2$	0,26	2,6	90°	0	2,6
$\vec{F}_3$	0.300	0,3 × 10 = 3N	240°	-1,5	-2,6

Q4: Newton's Second Law

In the Newton's second law experiment, if the traveled distance (s) during a time (t) is (s = 0.8m).
a) Complete the following table:

Hanging mass m_h (kg)	Time t (s)	Force (N)	Acceleration a (m/s^2)	Total mass (kg)
0.02	1.40	0.2	0.8	0,245
0.04	0.98	0.4	1.66	0,24
0.06	0.8	0.6	2.5	0,241

6

b) Write your conclusion?

when the mass increase the force increase too, but time decrease.

$a = \frac{2s}{t^2}$, $F = m_h g$

Q5: Uniform Circular Motion

In the circular motion experiment, a student studied the relation between the centripetal force and the square of the speed (v^2) for rotational body (**m**) by making the radius (**r=0.45 m**).

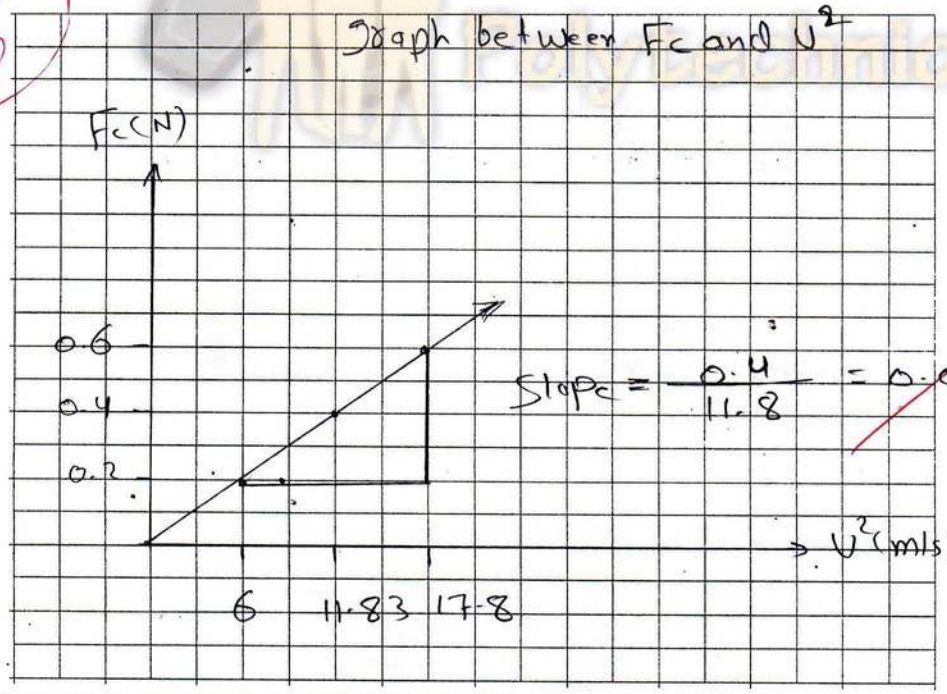
M (kg)	Period T (s)	F_c (N)	v(m/s)	v^2 $(m/s)^2$
0.02	1.15	0.2	2.45	6
0.04	0.82	0.4	3.448	11.83
0.06	0.67	0.6	4.219	17.8

a) Complete the table?

b) Plot a suitable graph between F_c and v^2

c) By using the slope of the graph, find the mass (m) of the rotational body.

6

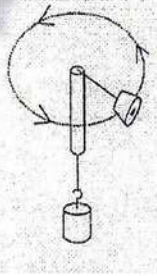


$Slope = \frac{0.4}{11.8} = 0.033\ N/m/s^2$

slope $= 4\pi^2 r \times m$

$m = \frac{slope}{4\pi^2 r}$

$m = 0.03\ kg$



Balqa Applied University
Faculty of Science
Department of Physics

13
20

Physics lab 111
Mid Exam

Name (بالعربي) ... Reg. Number: ٣٢٥٠٧١٧٨٠٣٣ Date: 30/06/2016

Class Number: ..... Instructor Name: ... Class Date & Time: ...

*MEASUREMENT AND UNCERTAINTIES:

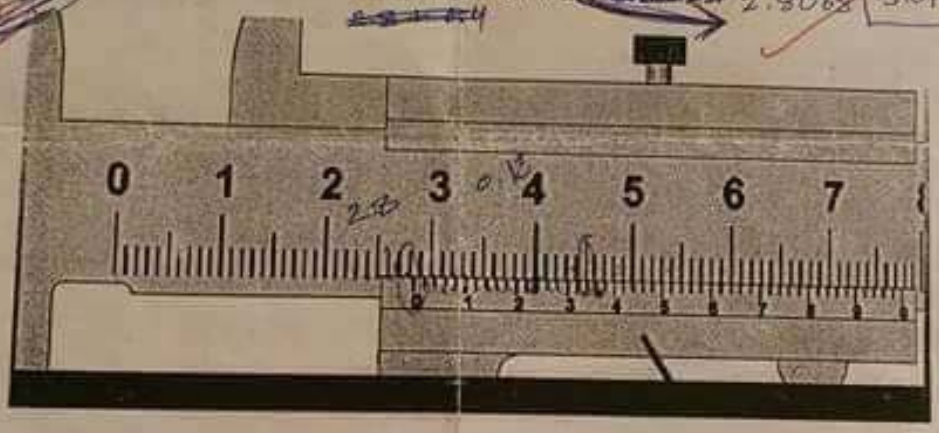
1) Fill in the blanks with the correct word or sentence: (6 marks)

(a) The error in computing the volume of cylinder ΔV Using the formula $V = \pi r^2 L$ is given by $\Delta V = V\sqrt{(2\frac{\Delta r}{r})^2 + (\frac{\Delta L}{L})^2}$

(b) The standard error of measurement (SEM) for the following lengths reading (4.5, 4.4, 4.3, 4.6) is: 0.12806

(c) If all of the scale lines of the main scale and vernier scale are all in alignment what is the reading of the vernier scale : the angel between them be = zero and slop$_1$ = slop$_2$

(d) The reading of the varnier calibre shown below is: 2.806 cm



*KINEMATIC OF RECTILINEAR MOTION:

2) Use the tape below to fill the table and to answer the questions below.(5 marks)



Time (s)	Position (cm)	Average Velocity (cm/s)	Average Acceleration (cm/s^2)
0.1	25		
0.15		270	
0.2	52		300
0.25		300	
0.3	82		

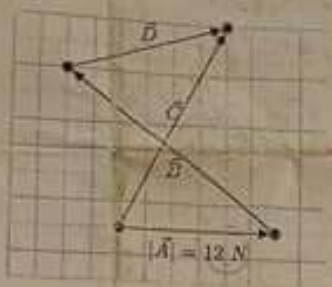
50 dot → 1s
5dot → 0.1s
Δt = 0.1 s

(a) The velocity during the time interval $t \in [0.3, 0.5]$ sec is:.....(Decreasing, Increasing, Constant).

(b) The acceleration during the time interval $t \in [0.2, 0.5]$ sec is:.....(Negative, Zero, Positive)

*VECTORS:

3) The figure below show the vector polygon method (graphical method) applied to find the resultant vector. Use the figure to answer the following (5 marks)



(a) Which of the above vectors represent the resultant vector: C

(b) The two components of the resultant vector are: R_x = 12.5 N R_y = 13.04 N

(c) If instead we use the force table, what would the the balance force angle be ... 62°

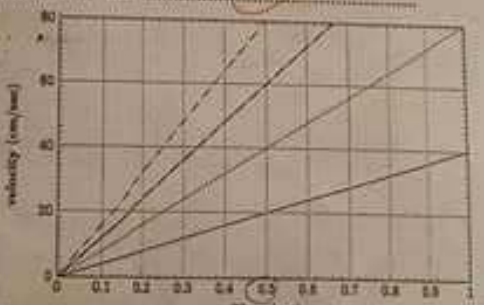
*FORCE AND MOTION:

4) Answer the following questions concerning the force and motion experiment: (5 marks)

(a) What is the objectives of this experiment: find the magnetud of the (v, a) and find the mtotal and mc, mh, mg

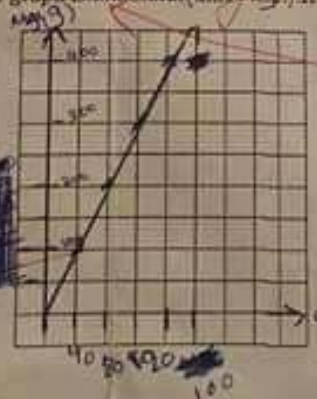
(b) Use the adjacent figure to fill the following table:

Added mass m_c (g)	Acceleration a (cm/s²)
100	40
200	80
300	120
400	160



(c) Plot a graph between the added mass (y-axis) and the acceleration (x-axis).

(d) The slope of the line of your graph is: mtotal (unit = g) It represent: 2.5 g



Slope = $\frac{200 - 100}{80 - 40}$

= $\frac{100}{40}$ = 2.5

$m_a + m_h + m_c = m_{total}$

0.5 0.5

$m_c = m_{total} - 0.5$

2.5 - 0.5 = 2

GOOD LUCK

Al- Balqa' Applied University
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Department of Physics and Basic Sciences

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم الفيزياء والعلوم الأساسية

Physics 111
Midterm Exam (a)
Time: 50 minutes

الاسم بالعربية: سلطان ابو شنب مدرس المقرر: د. زياد عيان موعد المقرر: ٢-٥

Circle the correct answer

1. The number of significant figures in the measurement (L = 75.00 m) is:

a) 1 b) 2 c) 3 d) 4

2. The addition of the measurements (6.201 cm, 7.4 cm, 0.68 cm, and 12.0 cm) equals

a) 26.281 cm b) 26.3 cm c) 26.28 cm d) 26.0 cm

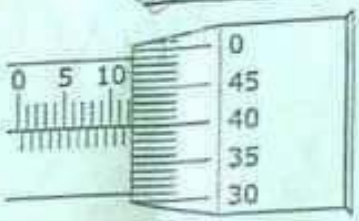
3. The multiplication of the measurements (3.2145 km × 4.23 km) equals

a) 13.6 km^2 b) 13.597335 km^2 c) 13.59 km^2 d) 13.60 km^2

4. Which of the following measurements represent the reading of the micrometer?

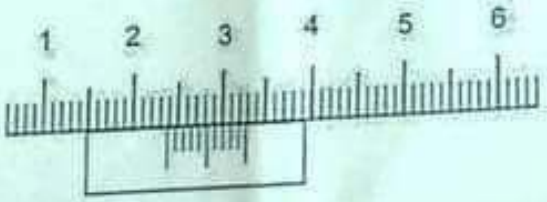
a) 10.3 mm b) 8.0 cm c) 1.50 cm d) 2.32 mm

5. The reading of the micrometer is:



a) 12.90 mm b) 10.240 mm
c) 12.40 mm d) 10.60 mm

6. The reading of the Vernier caliper is:

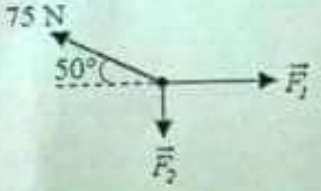


a) 2.30 cm b) 2.38 cm
c) 1.48 cm d) 2.48 cm

7. A cylinder has a radius of $R \pm \Delta R = (2.00 \pm 0.01)$ cm and a length of $L \pm \Delta L = (10.0 \pm 0.1)$ cm. The error in the volume ΔV is:

a) 0.02 cm^3 b) 0.015 cm^3 c) 2.51 cm^3 d) 1.9 cm^3

8. A stationary object has three forces acting on it, as shown in the diagram, What are the magnitude of $\vec{F}_1$?



a) 57.5 N b) 48.2 N c) 15 N d) -20 N

AL-BALQA' APPLIED UNIVERSITY
Faculty of Engineering Technology
Physics Department

Fall 2016
Date: April 7, 2016

Midterm
Physics 111

Name: _____ Instructor: _____

Consider $g = 10\ m/s^2$

1) Circle the correct the best answer for the following questions.

Q.1 A particle of mass 5 Kg is just starting to move when a force of 30 N is applied horizontally. What is the coefficient of static friction μ_s?

a) 0.3 b) 1.5 c) 0.7 d) 0.5 e) 0.6

Q2 A student wanted to calculate the density (ρ) of Mass (M). If $M \pm \Delta M = (10 \pm 0.01)$ gram and its volume $V \pm \Delta V = (5 \pm 0.1)\ cm^3$, the absolute error of the density ($\Delta\rho$) in g/cm^3 is:

a) 0.08 b) 0.04 c) 0.06 d) 0.02 e) none

Q.3 A Vernier caliper was used to measure the length of a cube. What is the error in length (ΔL) ?

a) 0.50 mm b) 0.05 mm c) 0.5 mm d) 0.005mm e) 0.0005 mm

Q.4 Which of the following measurements represents the reading of the micrometer?

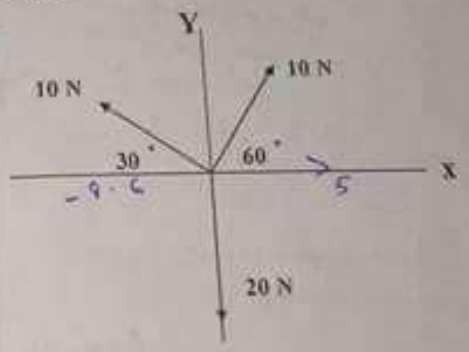
a) 9.25 mm b) 8.0 cm c) 1.50 cm d) 2 mm e) 10.1 mm

Q.5 What is the value of 0.484×0.080 to the appropriate number of significant figures?

a) 0.04 b) 0.03 c) 0.039 d) 0.0387 e) 0.038

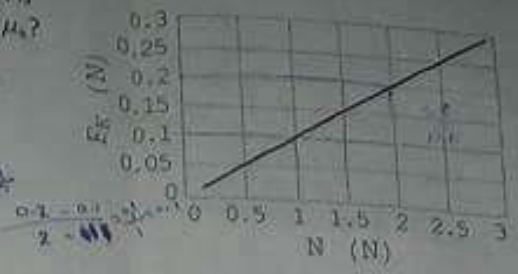
Q.6 In the Figure shown.
What is the magnitude of the resultant force (in N) along x axis?

a) 13.66 b) 5.00 c) 8.66 d) 3.66 e) 0



Q.7 From the graph shown. What is the coefficient of kinetic friction μ_k?

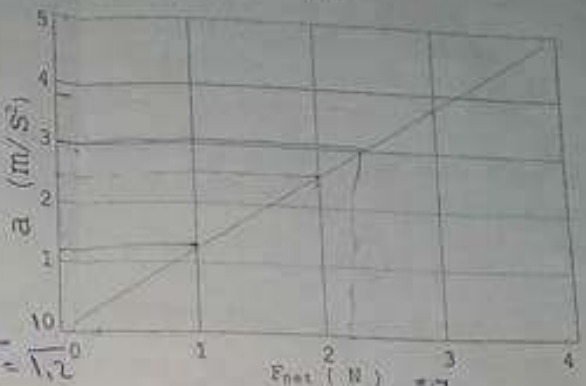
a) 0.1 b) 0.3 c) 0.4 d) 0.5
e) 0.7



2

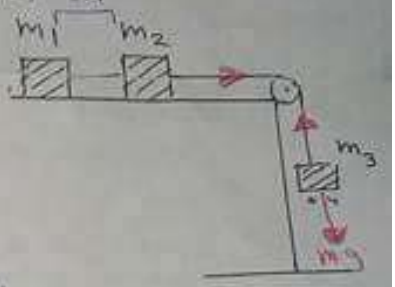
In the Newton's second law experiment, we add a new mass m_3, connected to m_2 as shown in the figure. (Show your calculations and write the final answer in the box).

a) What is the sum of the three masses.



Σm = Slope $= \frac{\Delta a}{\Delta a} = \frac{2-1}{2.6-1.4} = 1.2$

12 | 0.8 Kg |



b) Find F_{net} when $m_3 = 0.4$ Kg

$m_1 + m_2 = m_3$

$F = m g = 0.4 \times 9.8 = 3.92$ N

| 3.92 N |

c) What is the acceleration of the system when $m_3 = 0.3$ Kg?

$F = ma$

$3.92 = 0.3\,a$ $\quad a = 13.06$

| 13.0 m/s^2 |

$a = 3.75\ m/s^2$